

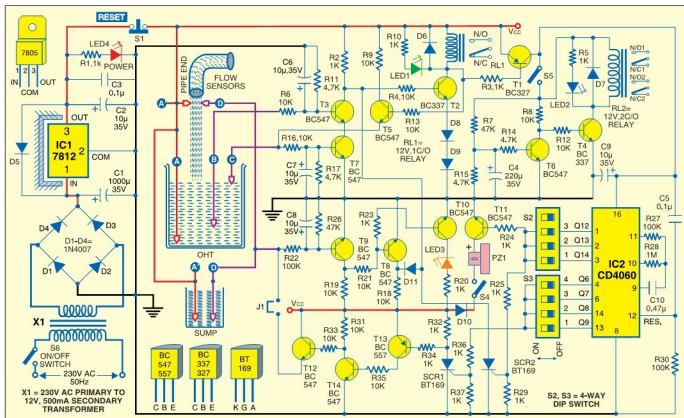


Fig. 2: Circuit of automatic water level controller for sump and submersible pump motors

level sensor B.

When the water level falls below sensor B, the water contact between sensors A and B is broken and transistor T3 goes into cut-off. Transistor T2 is driven into saturation to start the pump again.

Dry sump/no-flow sense and delay circuit. The dry sump/no-flow sense circuit consists of two npn transistors—T8 and T9. The sump water sensors A' and D, when in contact with water, provide positive bias to transistor T9, which goes into saturation sending transistor T8 into cut-off. The collector of transistor T8 is connected to the base of transistor T2 through diode D11. Cut-off state of transistor T8 does not affect the functioning of transistor T2. However, when there is no water in the sump, transistor T9 is cut-off, driving transistor T8 into saturation. Its collector becomes low and diode D11 provides negative bias voltage at the base of transistor T2 to switch it off. Thus the pump motor is stopped due to

dry sump conditions. This protects the pump motor from dry run.

In case of submersible pumps, flow sensors are used. These are fitted in OHT such that the inflowing water comes in contact with sensors before falling into OHT. It is observed that after starting the pump motor the inflow of water starts after time delay. This delay depends upon the depth of the bore well. The pump does not start till the flow sensors are in contact with water. The no-flow delay circuit simulates flow condition for a selected time so that the pump starts and runs for selected time period. In case the bore well has gone dry, or no water inflow takes place within the selected time, the circuit comes into action and stops the pump protecting it from dry run.

The no-flow delay (false flow simulator) circuit consists of timer IC CD4060 (IC2), DIP switch S3, SCR1, and transistors T13, T14 and T12.

Transistor T1 is switched on when transistor T2 is saturated. It provides

approximately 12V supply to start timer CD4060. Capacitor C10 with resistors R27 and R28, generates 30-, 60-, 120- and 240-second time-delay outputs at pin 4, 6, 14 and 13, respectively. DIP switch S3 facilitates selection of time delay. The output of timer activates SCR1 at selected time. Initially, SCR1 is off. Resistors R32 and R34 cut off transistor T13. Its collector is connected to the base of transistor T14, which is also cut off. Resistors R31 and R33 provide positive bias to transistor T12 driving it into saturation.

The emitter of transistor T12 is connected to sensor D through a shorting jumper J1. This simulates the inflow of water for selected period. If the inflow of water in OHT starts within the selected period, the pump continues to run. But if due to dry bore well or other reasons the inflow does not take place, the timer output turns on SCR1 and transistors T13 and T14 conduct. As a result, transistor T12 cuts off, removing the positive bias to transi-